Bus and decoder



File: bus.kicad_sch

IDE



File: ide.kicad_sch

Floppy and ROM



File: fdc.kicad_sch

Serial and parallel



File: io.kicad_sch

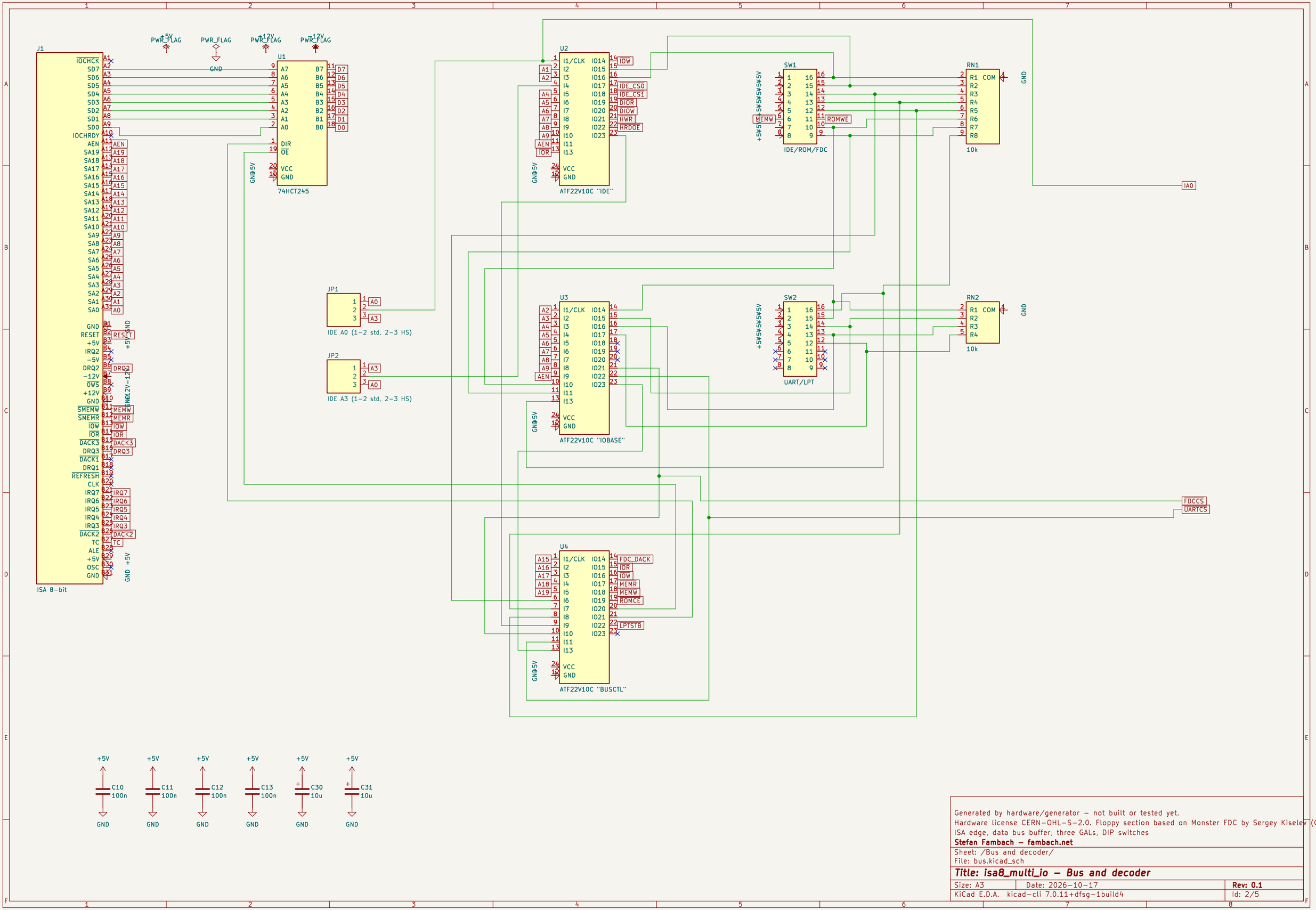
8-bit ISA multi I/O card: floppy (82077AA), XT-IDE compatible IDE, 16C550 serial, bidirectional parallel, 28C256 option ROM.
Glue logic in three ATF22V10C GALs (sources in gal/). Every block can be left unpopulated and switched off by DIP switch.
DIP switches: ON = 1. Jumpers select IRQ, DMA and power. Nets shared between sheets are global labels.

Generated by hardware/generator – not built or tested yet.
Floppy section based on Monster FDC by Sergey Kiselev (CERN-OHL-S-2.0)
Hardware license CERN-OHL-S-2.0
Stefan Fambach – fambach.net

Sheet: /
File: isa8_multi_io.kicad_sch

Title: isa8_multi_io – 8-bit ISA multi I/O card

Size: A4	Date: 2026-10-17	**Rev: 0.1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/5



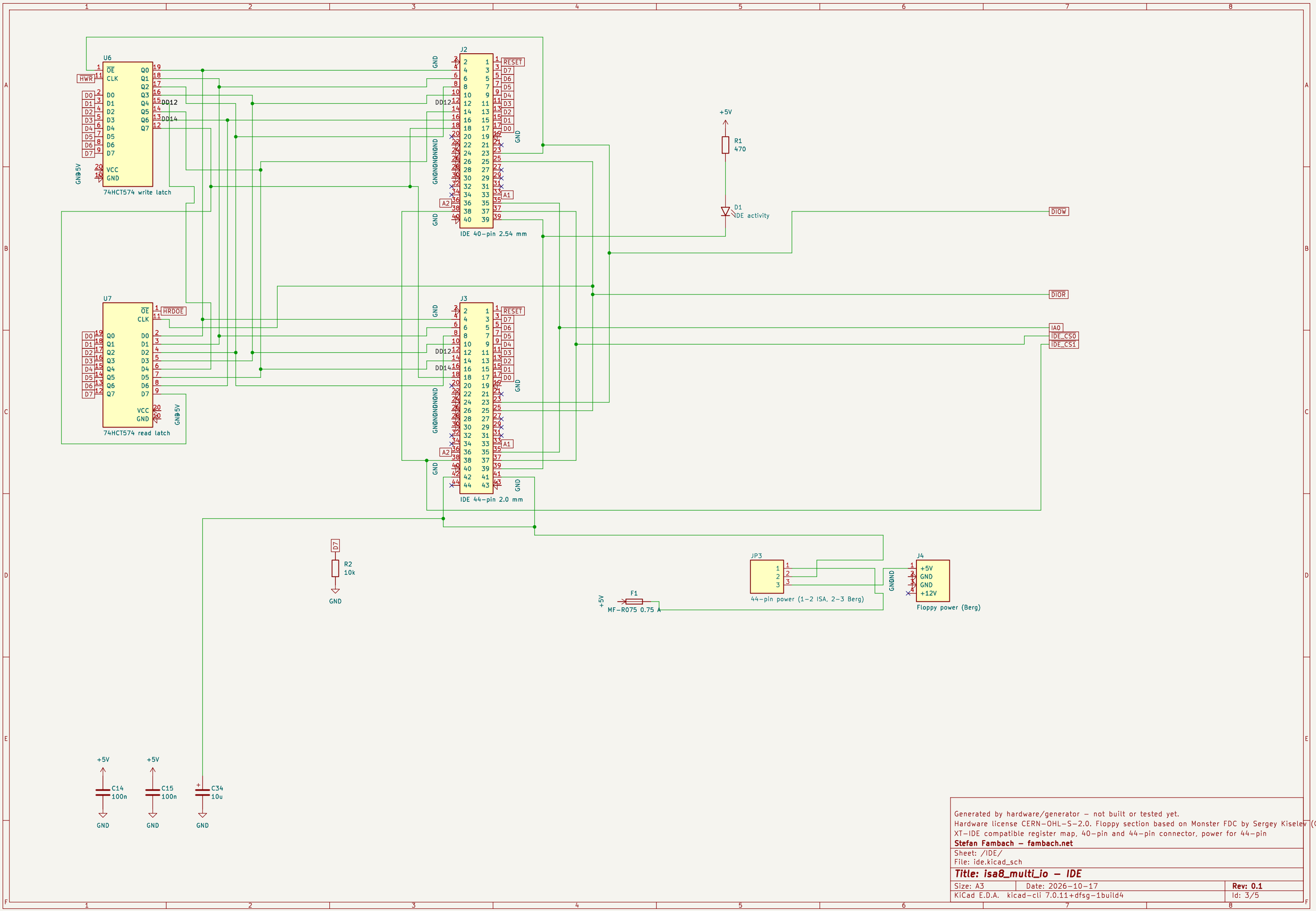
Generated by hardware/generator — not built or tested yet.
Hardware license CERN-OHL-S-2.0. Floppy section based on Monster FDC by Sergey Kiselev (CERN-O
ISA edge, data bus buffer, three GALs, DIP switches

Stefan Fambach — fambach.net

Sheet: /Bus and decoder/
File: bus.kicad_sch

Title: isa8_multi_io — Bus and decoder

Size: A3	Date: 2026-10-17	**Rev: 0.1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 2/5

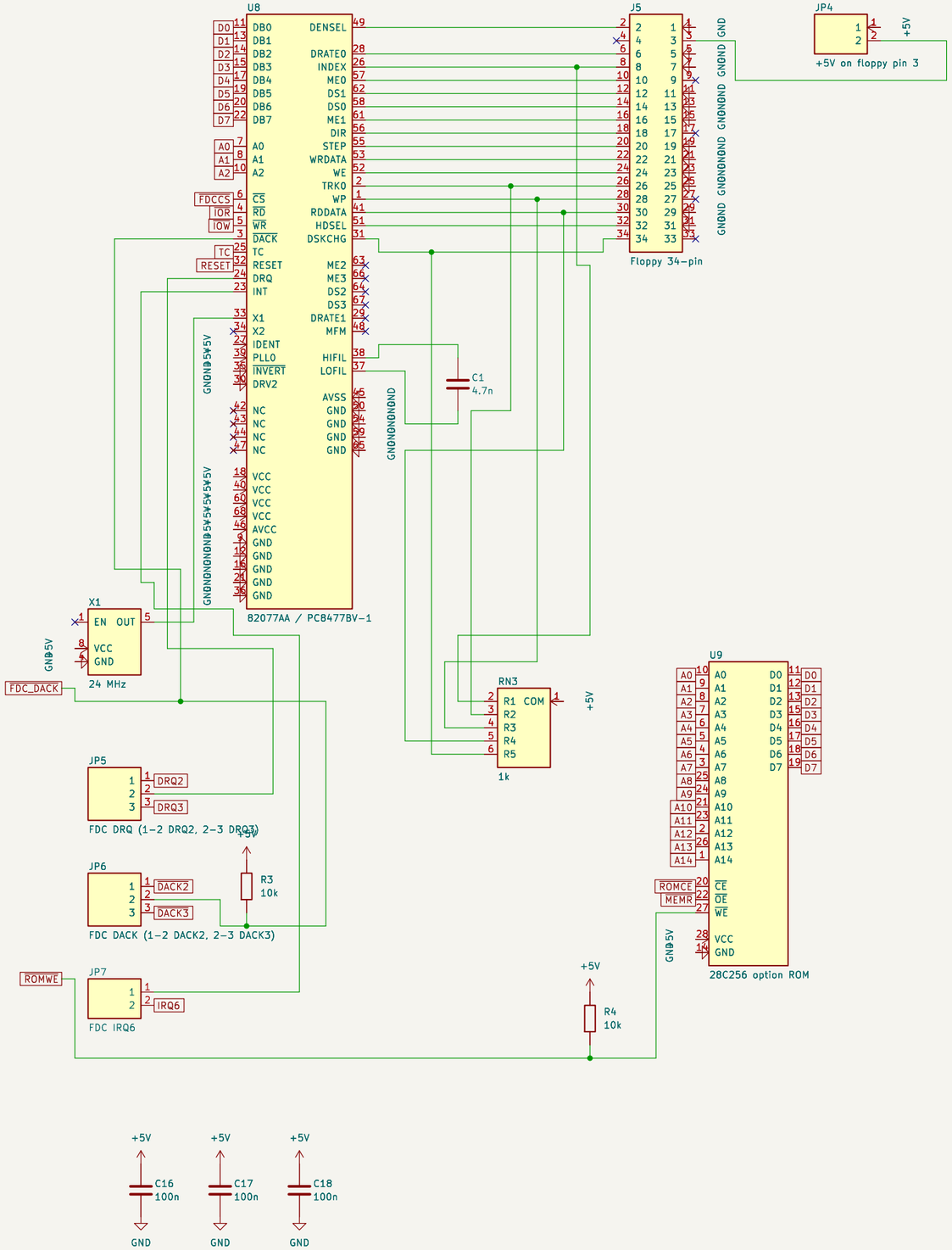


Generated by hardware/generator – not built or tested yet.
Hardware license CERN-OHL-S-2.0. Floppy section based on Monster FDC by Sergey Kiselev (CERN-O
XT-IDE compatible register map, 40-pin and 44-pin connector, power for 44-pin
Stefan Fambach – fambach.net

Sheet: /IDE/
File: ide.kicad_sch

Title: isa8_multi_io – IDE

Size: A3	Date: 2026-10-17	**Rev: 0.1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 3/5



Generated by hardware/generator – not built or tested yet.
Hardware license CERN-OHL-S-2.0. Floppy section based on Monster FDC by Sergey Kiselev (CERN-O
Floppy controller after Monster FDC, option ROM 28C256
Stefan Fambach – fambach.net

Sheet: /Floppy and ROM/
File: fdc.kicad_sch

Title: isa8_multi_io – Floppy and ROM

Size: A3	Date: 2026-10-17	**Rev: 0.1**
KiCad E.D.A.	kicad-cli 7.0.11+dfsg-1build4	Id: 4/5

X2
1.8432 MHz
U10
16C550
U11
SN75C185
J6
Serial DE9 male
U12
74HCT125
JP8
Serial IRQ 3/4/5/7
JP9
LPT IRQ (1–2 IRQ7, 2–3 IRQ5)
U19
74HCT138
U17
74HCT174 control
U18
7406
RN4
4k7
U5
74HCT04
U13
74HCT374 data latch
U14
74HCT244 data read
J7
Parallel DB25 female
U16
74HCT244 control read
U15
74HCT244 status read
C19 100n
C20 100n
C21 100n
C22 100n
C23 100n
C24 100n
C25 100n
C26 100n
C27 100n
C28 100n
C29 100n
C32 100n
C33 100n

Generated by hardware/generator – not built or tested yet.
Hardware license CERN-OHL-S-2.0. Floppy section based on Monster FDC by Sergey Kiselev (CERN-O
16C550 serial port, bidirectional parallel port
Stefan Fambach – fambach.net
Sheet: /Serial and parallel/
File: io.kicad_sch

Title: isa8_multi_io – Serial and parallel

Size: A2	Date: 2026-10-17	**Rev: 0.1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 5/5